EXPERIMENTAL PAPER BLUEPRINT & RESEARCH GAP ANALYSIS

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Topic: A Machine Learning Framework for Predicting Temperature Anomalies and Global Warming Trends in Nepal using Earth Observation Data

1. Executive Synthesis & Comparative Literature Matrix

This blueprint synthesizes the three foundational base papers in Tripti.zip alongside our developed NASA POWER machine learning framework:
• Base Paper 1 (Nepal et al., 2025 — WeatherWave): Implements a 5-tree Random Forest regression model on daily NASA POWER satellite reanalysis across 77 districts of Nepal for 1-day persistence temperature forecasting (MAE = 0.424°C, RMSE = 0.695°C, R^2 = 0.9934). Research Gap: Restricted to 1-day persistence regression without multi-class thermal anomaly classification, unsupervised regime clustering, or multi-decadal 2050 global warming trajectory projections.
• Base Paper 2 (Bhattarai, 2022 — Analysis of Heat Wave Over Nepal, TU Dissertation): Analyzes daily maximum temperature from 46 DHM ground stations across 6 physiographic/basin regions (1991–2020) using a 90th-percentile + 4°C threshold for >=3 consecutive days in MATLAB/Excel. Research Gap: Relies solely on sparse ground stations (missing high-elevation coverage >3,000 m) and manual descriptive thresholding with zero predictive machine learning or feature importance attribution.
• Base Paper 3 (Sapkota, 2007 — Temporal and Spatial Variation of Temperature Over Nepal, TU Dissertation): Evaluates 1970–2004 maximum and minimum temperature trends across Nepal's 5 physiographic zones using linear OLS trend lines (national Tmax +0.08°C/yr, Tmin +0.05°C/yr; Jomsom high-altitude station cooling in Tmax but warming in Tmin). Research Gap: Pre-2004 dataset analyzed with static univariate linear regression; incapable of capturing non-linear diurnal thermal range (DTR) compression or discrete anomaly regimes.

2. How Our Developed Framework Resolves All Three Base Paper Gaps

• 27-Year High-Elevation NASA POWER MERRA-2 Archive (2000–2026): Ingests N = 9,763 continuous daily records at 28.30°N, 83.94°E (elevation 3,042.47 m in the central Himalayan corridor) with 0% missing dropouts.
• Non-Parametric Mann-Kendall & Sen's Slope Verification: Proves statistically significant nocturnal warming in Monsoon (+0.3017°C/decade, p = 0.0004) and Post-Monsoon (+0.9025°C/decade, p = 0.0001) alongside severe Diurnal Temperature Range (DTR) compression (-0.7038°C/decade, p < 0.0001).
• Unsupervised + Supervised + Continuous + 2050 Forecasting Suite: Integrates SVD-PCA (76.57% variance in 3 PCs), K-Means++ (k = 3 physical thermal regimes), 5 supervised classifiers (XGBoost 70.41% accuracy, OvR AUC = 0.8720; Random Forest 69.93%), continuous next-day temperature forecasting (XGBoost Regressor RMSE = 0.8474°C, R^2 = 0.9732 beating API

persistence RMSE = 0.9343°C), and IPCC SSP1-2.6 / SSP2-4.5 / SSP5-8.5 warming projections through 2050.